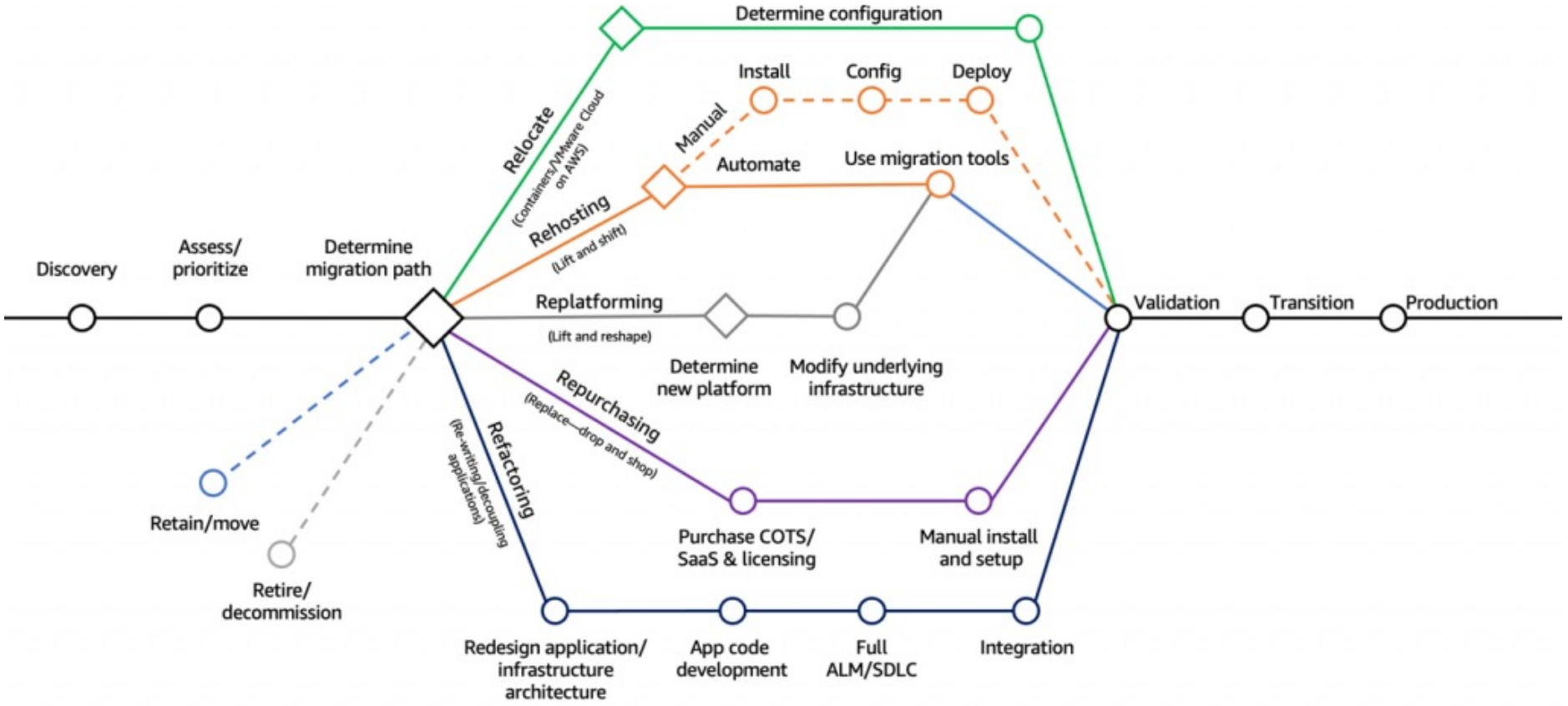
Migrations

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Cloud Migration Strategies: The 7Rs



https://aws.amazon.com/blogs/enterprise-strategy/new-possibilities-seven-strategies-to-accelerate-your-application-migration-to-aws/

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Cloud Migration Strategies: The 7Rs

- Retire
  - Turn off things you don't need (maybe as a result of Re-architecting)
  - Helps with reducing the surface areas for attacks (more security)
  - Save cost, maybe up to 10% to 20%
  - Focus your attention on resources that must be maintained

- Retain
  - Do nothing for now (it's still a decision to make in a Cloud Migration)
  - Security, data compliance, performance, unresolved dependencies
  - No business value to migrate, mainframe or mid-range and non-x86 Unix apps

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Cloud Migration Strategies: The 7Rs

- Relocate
  - Move apps from on-premises to its Cloud version
  - Move EC2 instances to a different VPC, AWS account or AWS Region
  - Example: transfer servers from VMware Software-defined Data Center (SSDC) to VMware Cloud on AWS

- Rehost *"lift and shift"*
  - Simple migrations by re-hosting on AWS (applications, databases, data…)
  - Migrate machines (physical, virtual, another Cloud) to AWS Cloud
  - No cloud optimizations being done, applications is migrated as is
  - Could save as much as 30% on cost
  - Example: Migrate using AWS Application Migration Service

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Cloud Migration Strategies: The 7Rs

- Replatform *"lift and reshape"*
  - Example: migrate your database to RDS
  - Example: migrate your application to Elastic Beanstalk
  - Not changing the core architecture, but leverage some Cloud optimizations
  - Save time and money by moving to a fully managed service or Serverless

- Repurchase *"drop and shop"*
  - Moving to a different product while moving to the Cloud
  - Often you move to a SaaS platform
  - Expensive in the short term, but quick to deploy
  - Example: CRM to Salesforce.com, HR to Workday, CMS to Drupal

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Cloud Migration Strategies: The 7Rs

- **Refactor / Re-architect**
  - Reimagining how the application is architected using Cloud Native features
  - Driven by the need of the business to add features and improve scalability, performance, security, and agility
  - Move from a monolithic application to micro-services
  - Example: move an application to Serverless architectures, use AWS S3

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS Storage Gateway



- Bridge between on-premises data and cloud data
- Use cases:
  - disaster recovery
  - backup & restore
  - tiered storage
  - on-premises cache & low-latency files access
- Types of Storage Gateway:
  - S3 File Gateway
  - Volume Gateway
  - Tape Gateway



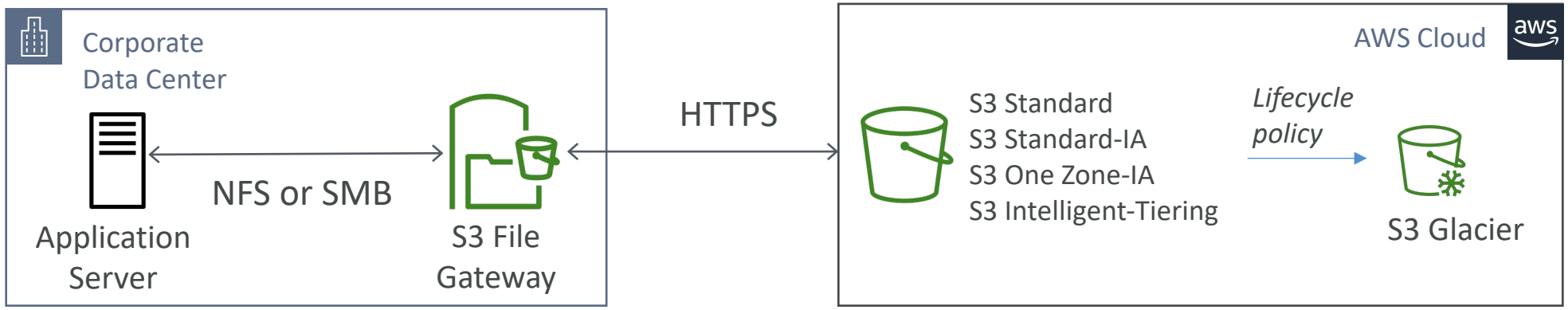
Storage Gateway

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Amazon S3 File Gateway

- Configured S3 buckets are accessible using the NFS and SMB protocol
- **Most recently used data is cached in the file gateway**
- Supports S3 Standard, S3 Standard IA, S3 One Zone A, S3 Intelligent Tiering
- **Transition to S3 Glacier using a Lifecycle Policy**
- Bucket access using IAM roles for each File Gateway
- SMB Protocol has integration with Active Directory (AD) for user authentication

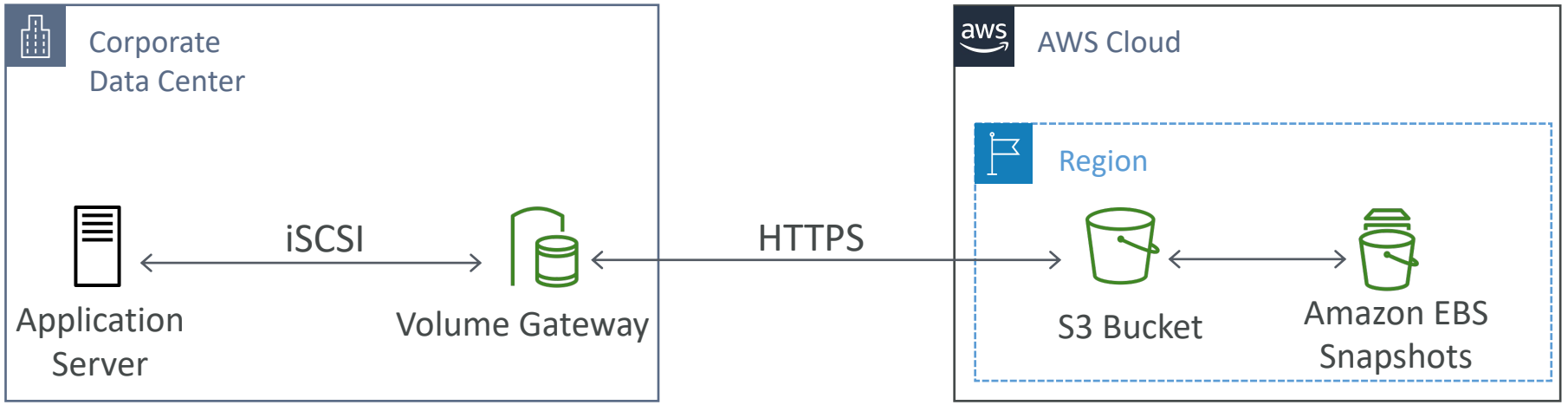


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Volume Gateway

- Block storage using iSCSI protocol backed by S3
- Backed by EBS snapshots which can help restore on-premises volumes!
- **Cached volumes:** low latency access to most recent data
- **Stored volumes:** entire dataset is on premise, scheduled backups to S3

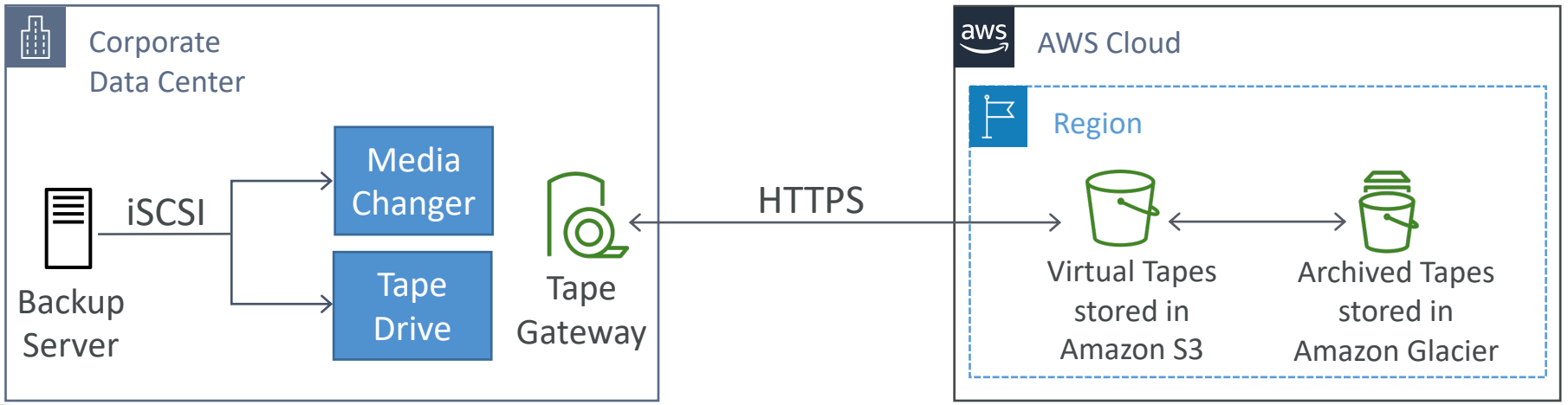


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Tape Gateway

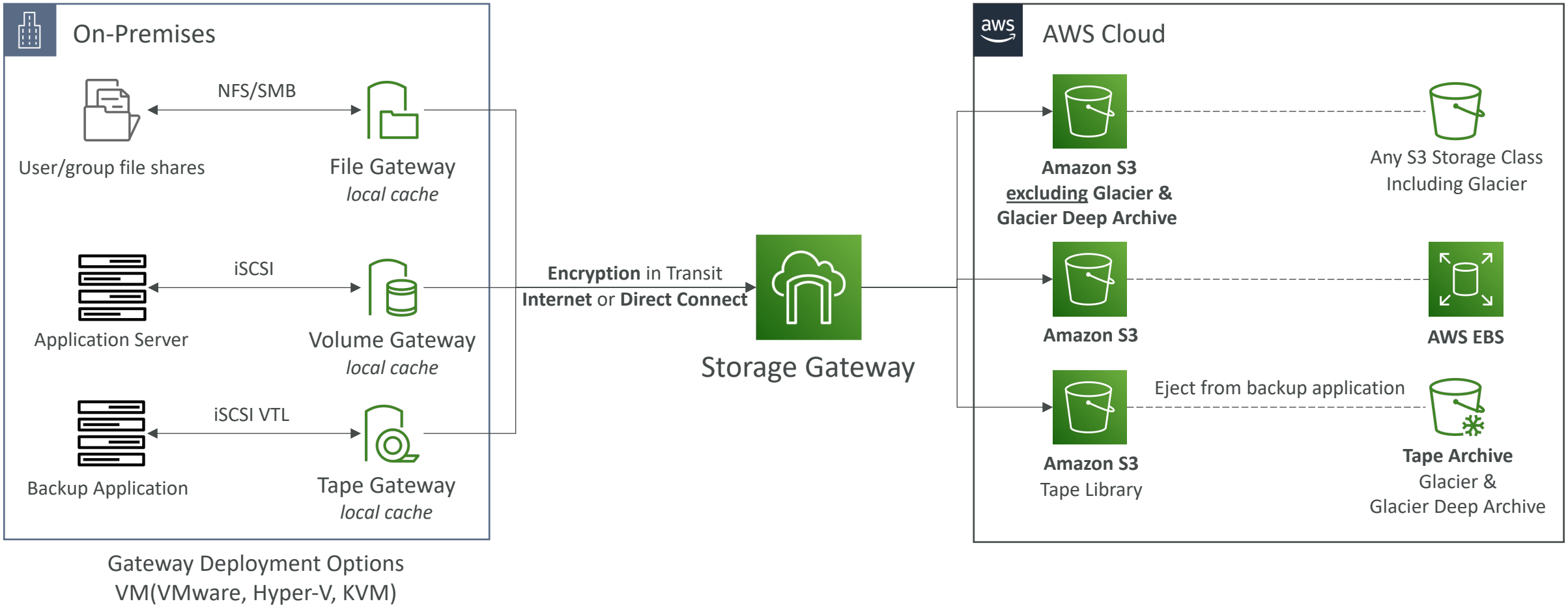
- Some companies have backup processes using physical tapes (!)
- With Tape Gateway, companies use the same processes but, in the cloud
- Virtual Tape Library (VTL) backed by Amazon S3 and Glacier
- Back up data using existing tape-based processes (and iSCSI interface)
- Works with leading backup software vendors



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS Storage Gateway

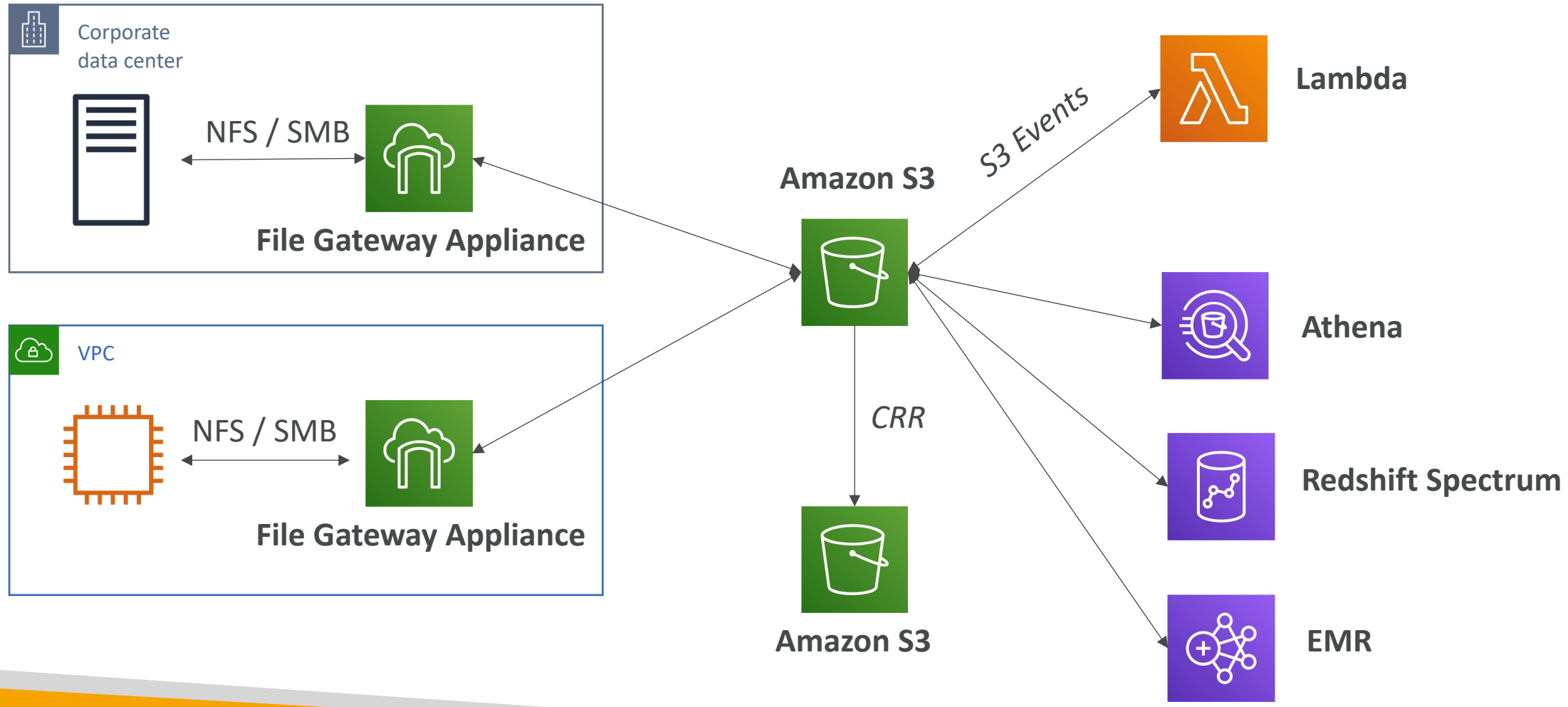


Gateway Deployment Options
VM(VMware, Hyper-V, KVM)

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

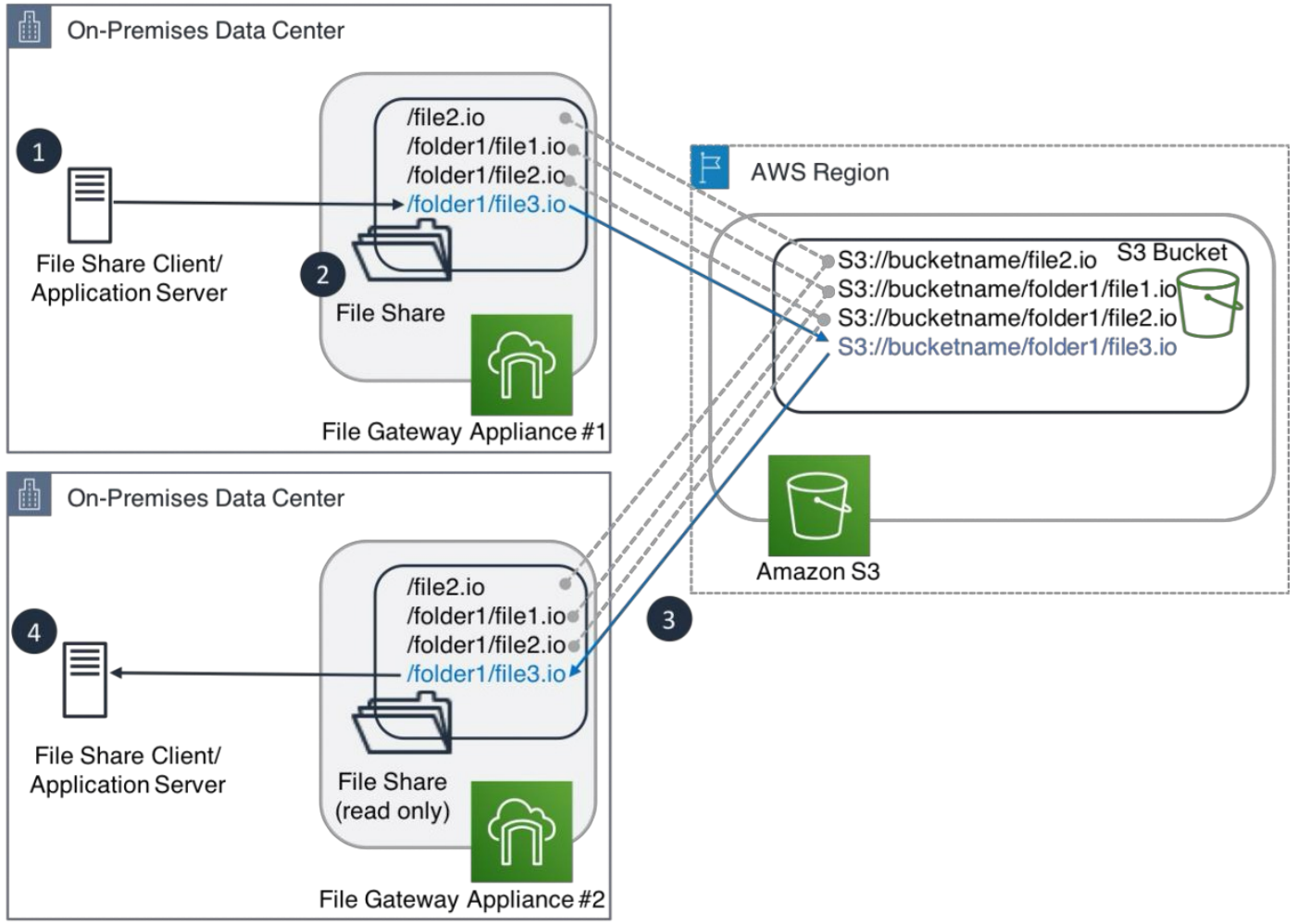
File Gateway: Extensions



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

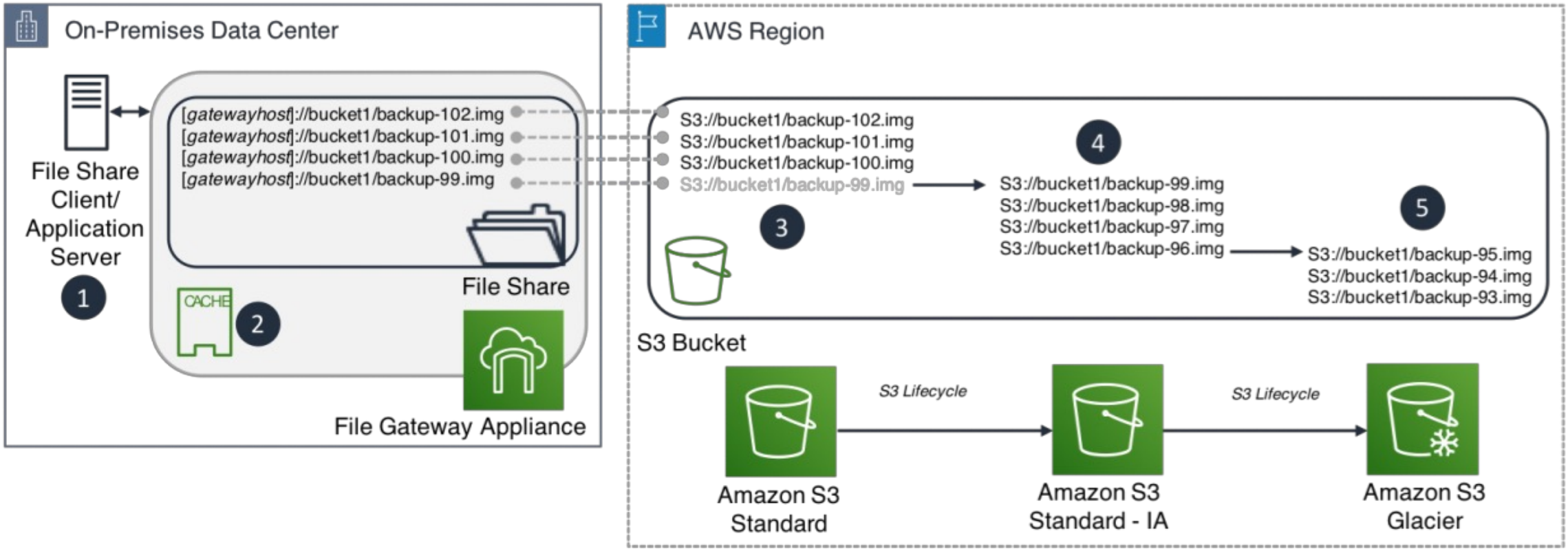
File Gateway: Read Only Replicas



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

File Gateway: Backup and Lifecycle Policies



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

File Architectures: Other possibilities

- Amazon S3 Object Versioning
  - Ability to store multiple object versions as they are modified
  - Helpful to restore a file to a previous version
  - Could restore an entire file system to a previous version
  - Must use the "RefreshCache" API on the Gateway to be notified of restore

- Amazon S3 Object Lock
  - Enables to have the File Gateway for Write Once Read Many (WORM) data
  - If there are file modifications or renames in the file share clients, the file gateway creates a new version of the object without affecting priori versions, and the original locked version will remain unchanged

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS Snowball



- Highly-secure, portable devices to **collect and process data at the edge**, and **migrate data into and out of AWS**
- Helps migrate up to Petabytes of data

Device	Compute	Memory	Storage (SSD)
Snowball Edge **Storage** Optimized	104 vCPUs	416 GB	**210 TB**
Snowball Edge **Compute** Optimized	104 vCPUs	416 GB	**28 TB**





Snowball Edge

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Data Migrations with Snowball

	Time to Transfer		
	100 Mbps	**1Gbps**	**10Gbps**
10 TB	12 days	30 hours	3 hours
100 TB	124 days	12 days	30 hours
1 PB	3 years	124 days	12 days

Challenges:

- Limited connectivity
- Limited bandwidth
- High network cost
- Shared bandwidth (can't maximize the line)
- Connection stability

AWS Snowball: offline devices to perform data migrations

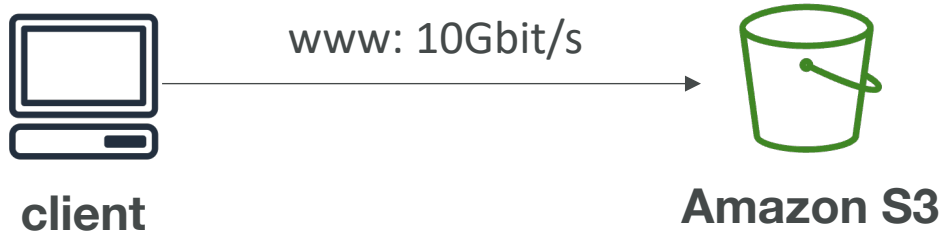
If it takes more than a week to transfer over the network, use Snowball devices!

© Stephane Maarek

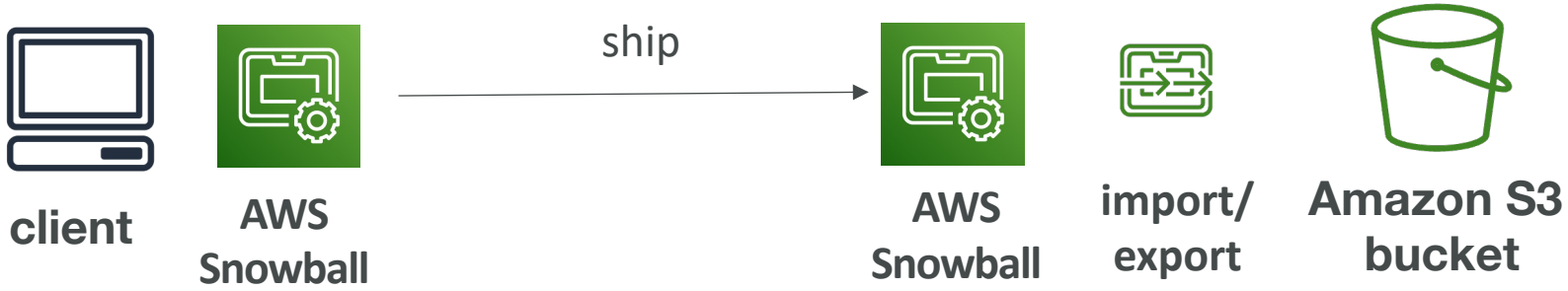
NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Diagrams

- Direct upload to S3:



- With Snowball:

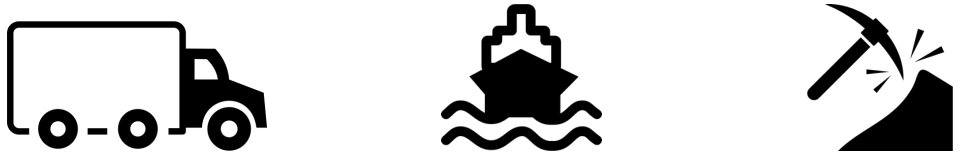


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

What is Edge Computing?

- Process data while it's being created on **an edge location**
  - A truck on the road, a ship on the sea, a mining station underground...



- These locations may have limited internet and no access to computing power
- We setup a **Snowball Edge** device to do edge computing
  - Snowball Edge Compute Optimized (dedicated for that use case) & Storage Optimized
  - Run EC2 Instances or Lambda functions at the edge
- Use cases: preprocess data, machine learning, transcoding media

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Snow Family – Improving Transfer Performance

- Most impactful to least:
  - **Perform multiple write operations at one time** - from multiple terminals
  - **Transfer small files in batches** – zip up small files until at least 1MB
  - **Don't perform other operations on files during transfer**
  - **Reduce local network use**
  - **Eliminate unnecessary hops** – directly connect to the computer

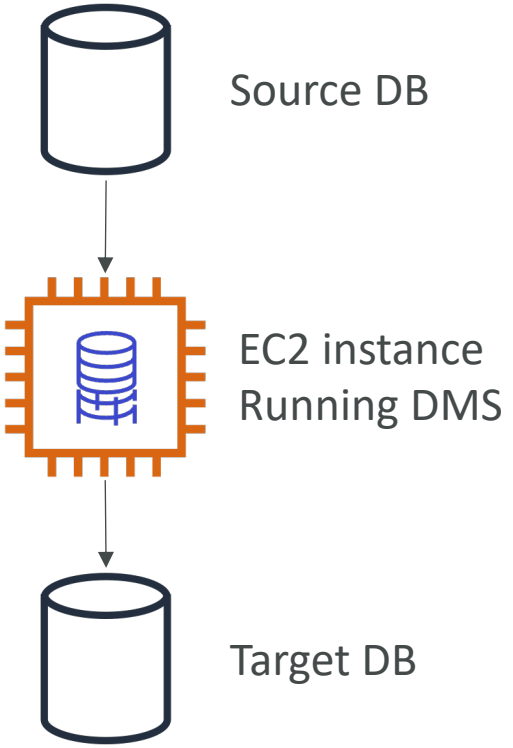
- The data transfer rate using the file interface is typically between 25 MB/s and 40 MB/s. If you need to transfer data faster than this, use the **Amazon S3 Adapter for Snowball**, which has a data transfer rate typically between 250 MB/s and 400 MB/s

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

DMS – Database Migration Service

- Quickly and securely migrate databases to AWS, resilient, self healing
- The source database remains available during the migration
- Supports:
  - Homogeneous migrations: ex Oracle to Oracle
  - Heterogeneous migrations: ex Microsoft SQL Server to Aurora
- Continuous Data Replication using CDC
- You must create an EC2 instance to perform the replication tasks



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

DMS Sources and Targets

SOURCES:

- On-premises and EC2 instances databases: *Oracle, MS SQL Server, MySQL, MariaDB, PostgreSQL, MongoDB, SAP, DB2*
- Azure: *Azure SQL Database*
- Amazon RDS: all including Aurora
- Amazon S3
- DocumentDB

TARGETS:

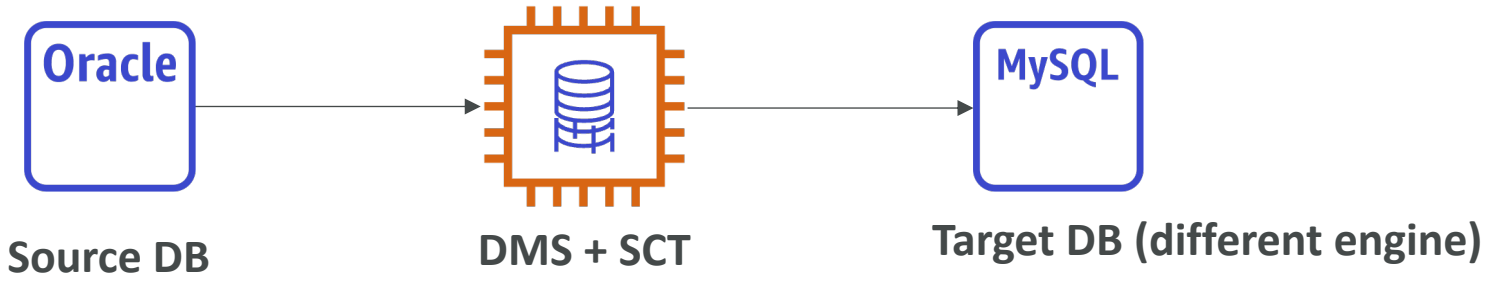
- On-premises and EC2 instances databases: Oracle, MS SQL Server, MySQL, MariaDB, PostgreSQL, SAP
- Amazon RDS including Aurora
- Amazon Redshift
- Amazon DynamoDB
- Amazon S3
- OpenSearch Service
- Kinesis Data Streams
- DocumentDB

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS Schema Conversion Tool (SCT)

- Convert your Database's Schema from one engine to another
- Example OLTP: (SQL Server or Oracle) to MySQL, PostgreSQL, Aurora
- Example OLAP: (Teradata or Oracle) to Amazon Redshift



- You do not need to use SCT if you are migrating the same DB engine
  - Ex: on-premises PostgreSQL => RDS PostgreSQL
  - The DB engine is still PostgreSQL (RDS is the platform)

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

DMS – Good things to know

- Works over VPC Peering, VPN (site to site, software), Direct Connect
- Supports Full Load, Full Load + CDC, or CDC only
- **Oracle:**
  - Source: Supports TDE for the source using "BinaryReader"
  - Target: Supports BLOBs in tables that have a primary key, and TDE
- **OpenSearch:**
  - Source: does not exist
  - Target: possible to migrate from a relational database using DMS
  - Therefore, DMS cannot be used to replicate OpenSearch data

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Snowball + Database Migration Service (DMS)

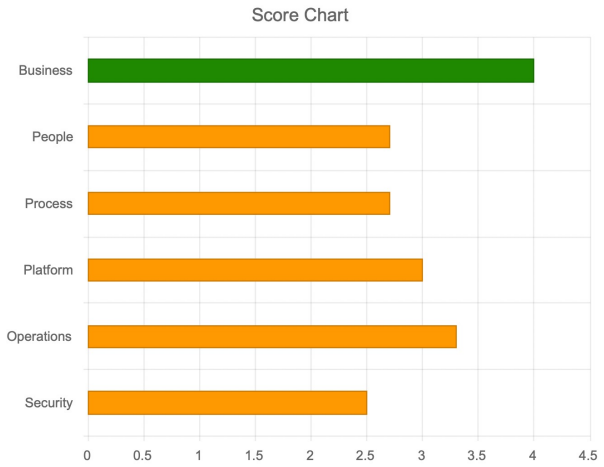
- Larger data migrations can include many terabytes of information.
- Can be limited due to network bandwidth or size of data
- AWS DMS can use Snowball Edge & Amazon S3 to speed up migration
- Following stages:
  1. You use the AWS Schema Conversion Tool (AWS SCT) to extract the data locally and move it to an Edge device.
  2. You ship the Edge device or devices back to AWS.
  3. After AWS receives your shipment, the Edge device automatically loads its data into an Amazon S3 bucket.
  4. AWS DMS takes the files and migrates the data to the target data store. If you are using change data capture (CDC), those updates are written to the Amazon S3 bucket and then applied to the target data store.

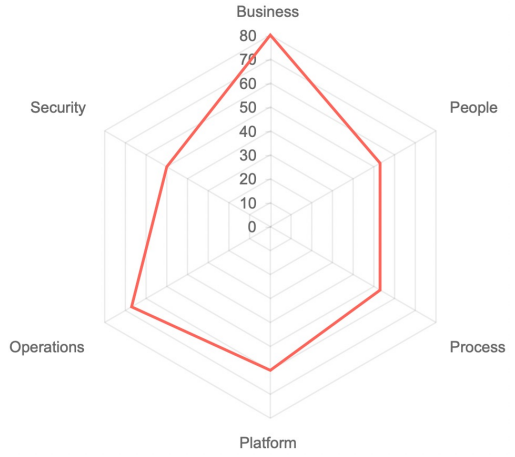
© Stephane Maarek

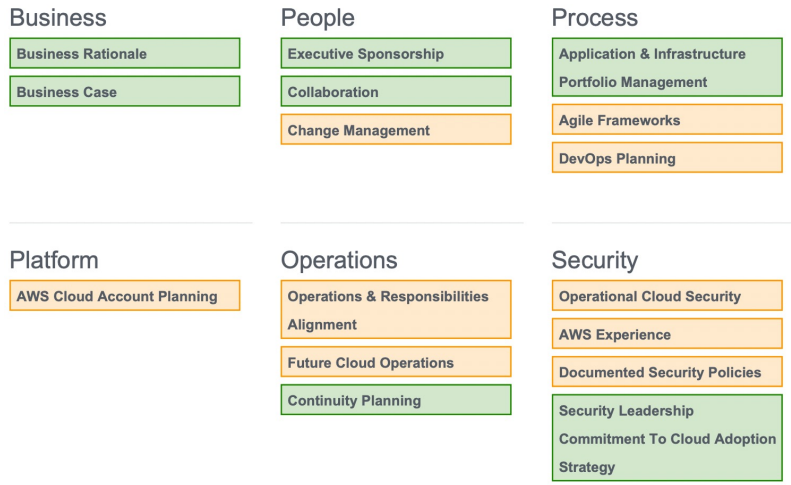
NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS Cloud Adoption Readiness Tool (CART)

- Helps organizations develop efficient and effective plans for cloud adoption and migrations
- Transforms your idea of moving to the cloud into a detailed plan that follows AWS best practices
- Answer a set of questions across six perspectives (business, people, process, platform, operations, security)
- Generates a custom report on your level of migration readiness







© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

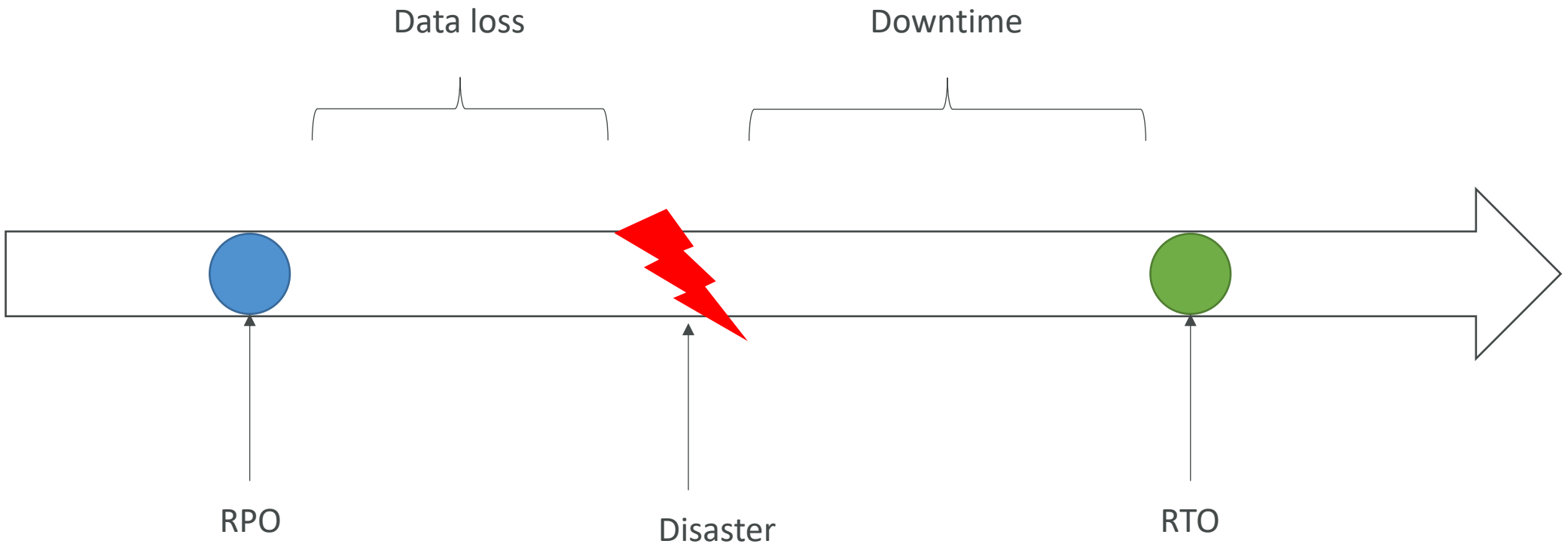
Disaster Recovery Overview

- Any event that has a negative impact on a company's business continuity or finances is a disaster
- Disaster recovery (DR) is about preparing for and recovering from a disaster
- What kind of disaster recovery?
  - on-premises => on-premises: traditional DR, and very expensive
  - on-premises => AWS Cloud: hybrid recovery
  - AWS Cloud Region A => AWS Cloud Region B
- Need to define two terms:
  - RPO: Recovery Point Objective
  - RTO: Recovery Time Objective

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

RPO and RTO

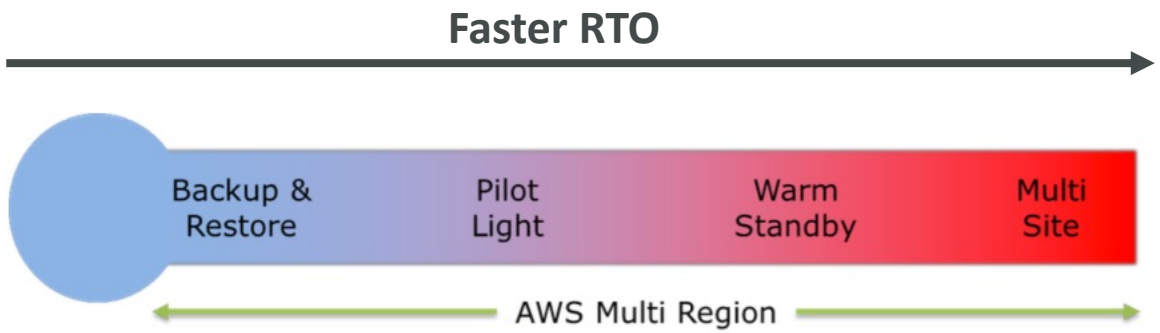


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Disaster Recovery Strategies

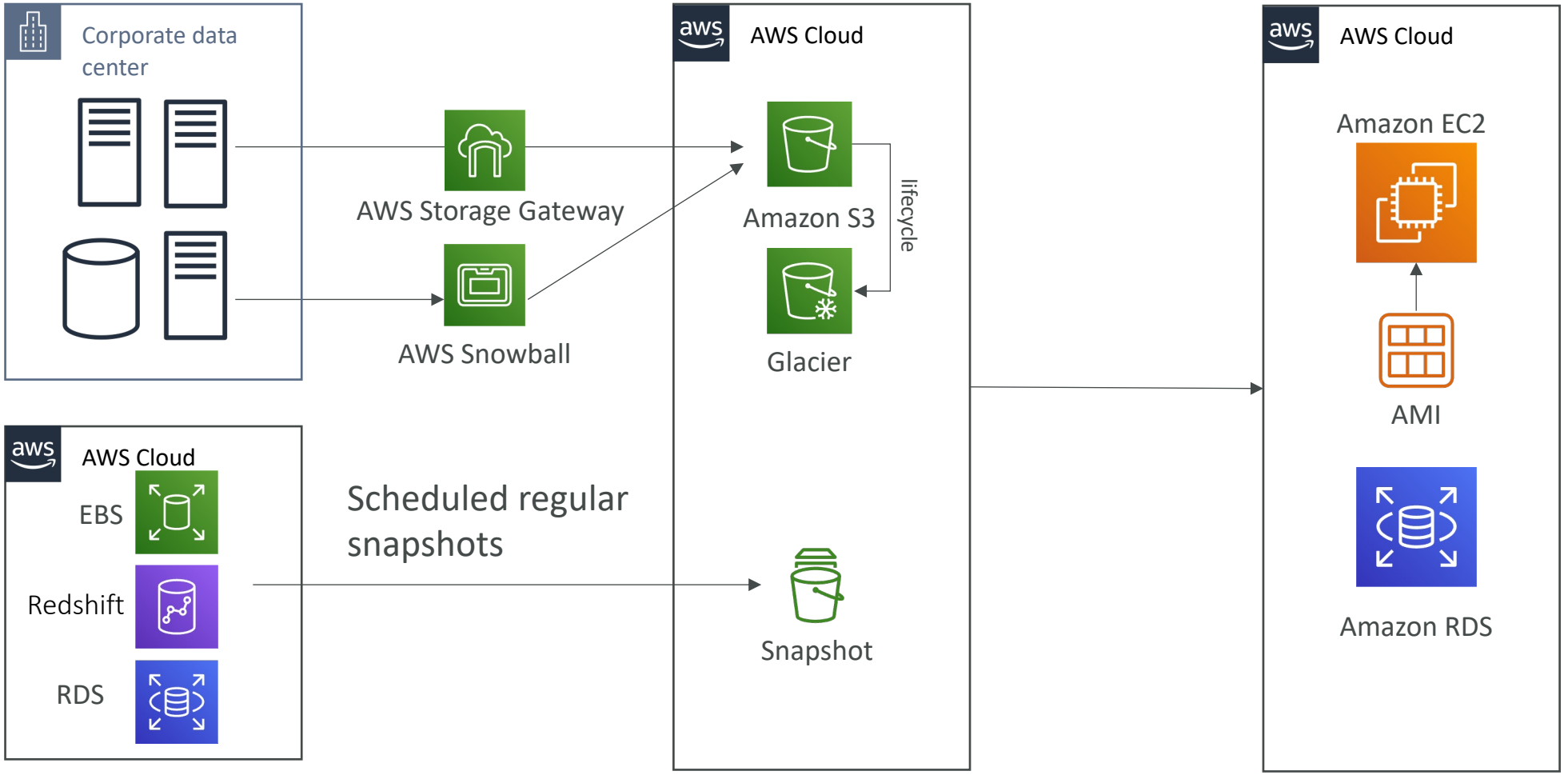
- Backup and Restore
- Pilot Light
- Warm Standby
- Hot Site / Multi Site Approach



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Backup and Restore (High RPO)

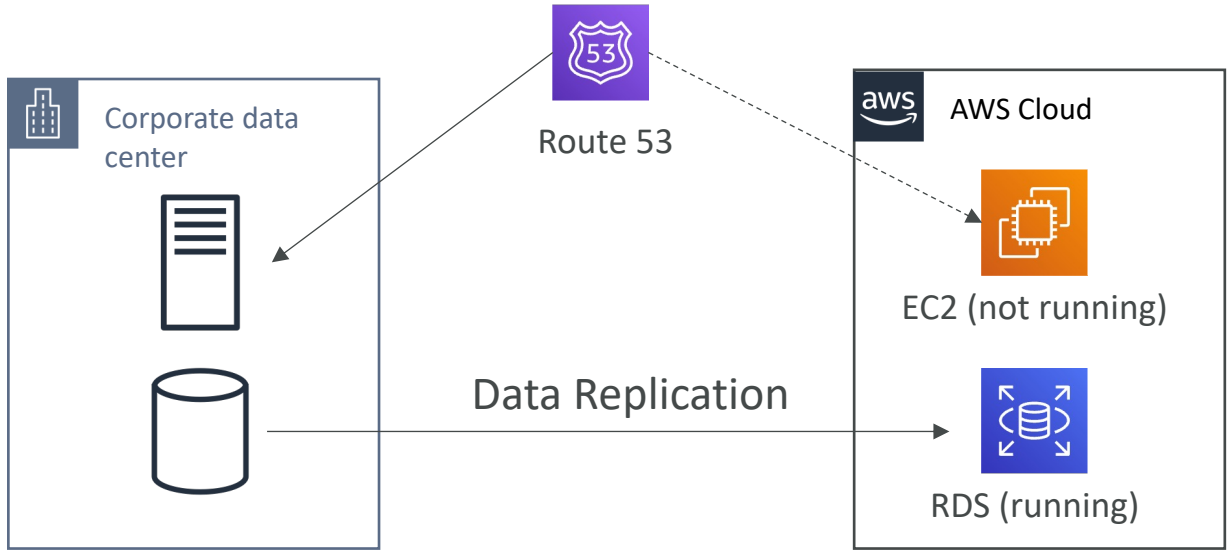


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Disaster Recovery – Pilot Light

- A small version of the app is always running in the cloud
- Useful for the critical core (pilot light)
- Very similar to Backup and Restore
- Faster than Backup and Restore as critical systems are already up

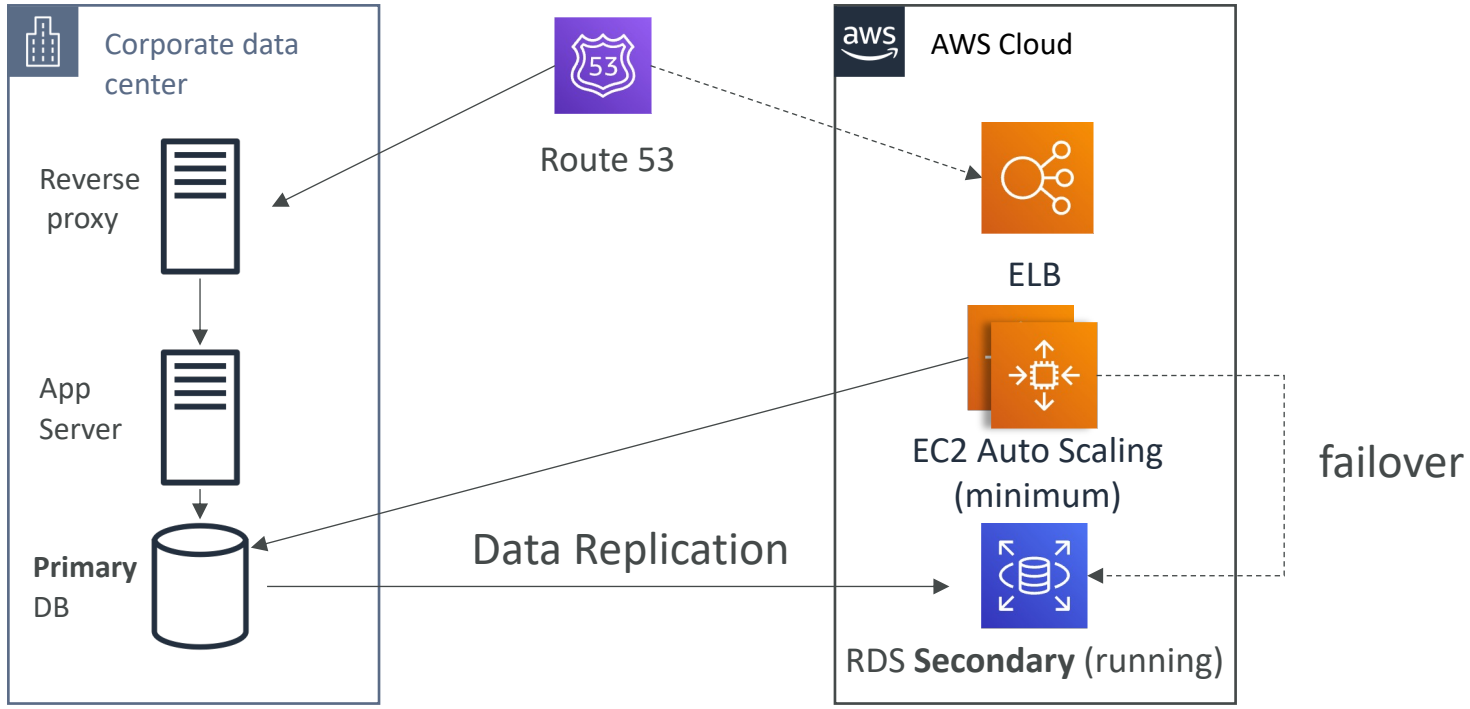


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Warm Standby

- Full system is up and running, but at minimum size
- Upon disaster, we can scale to production load

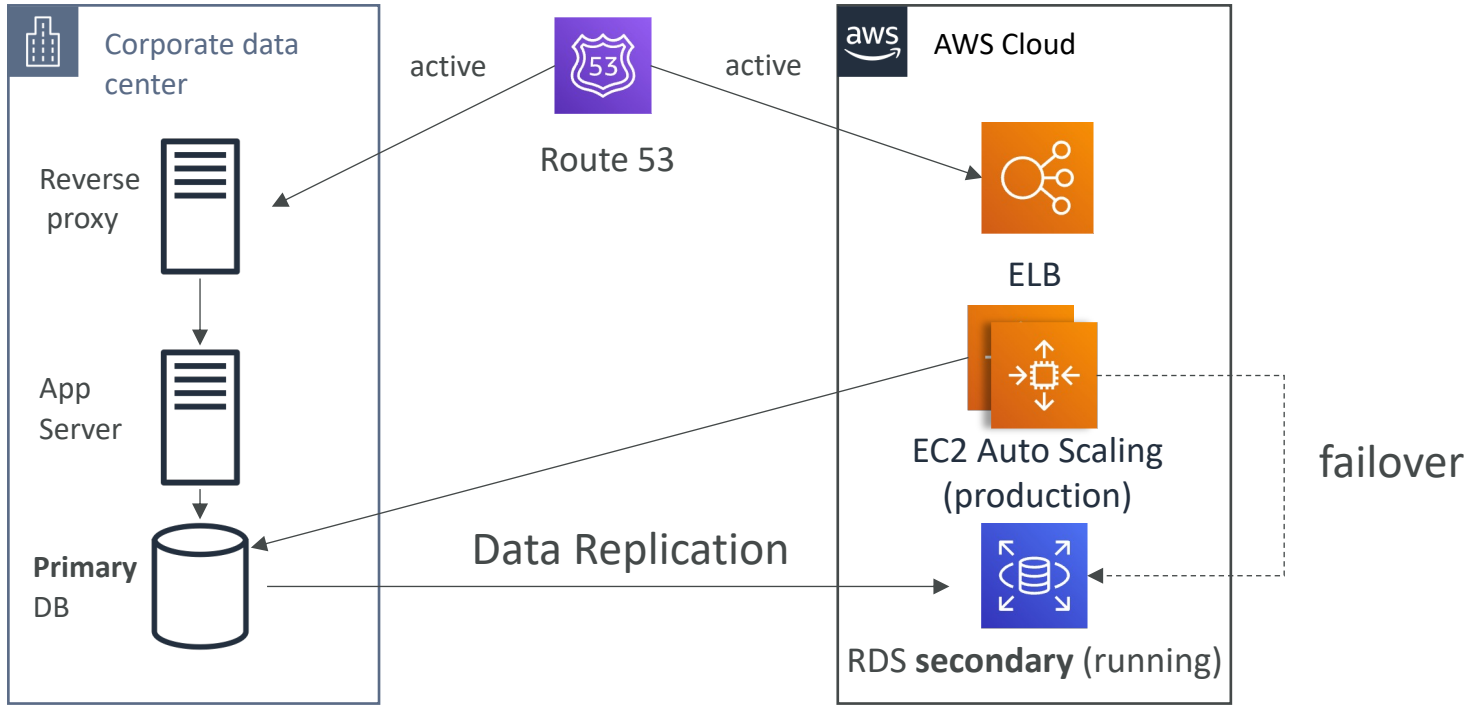


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Multi Site / Hot Site Approach

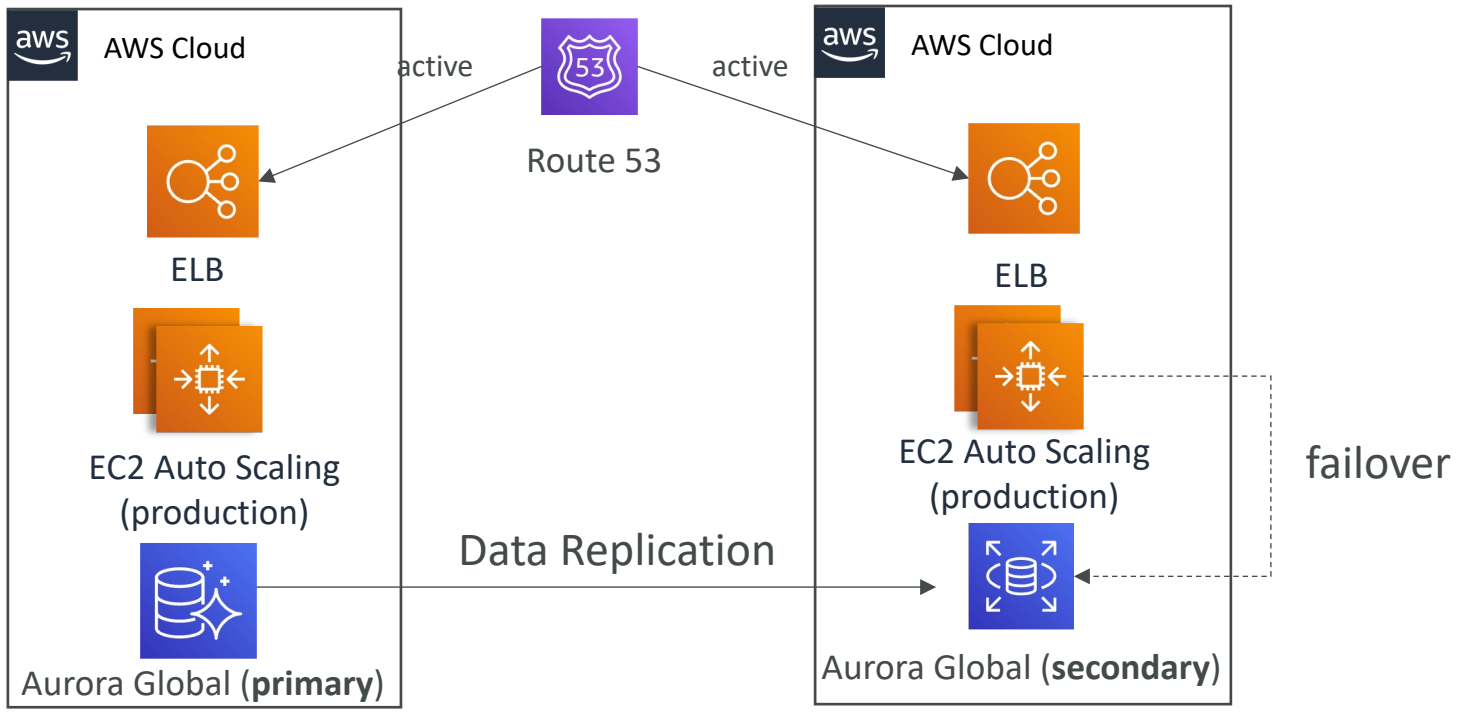
- Very low RTO (minutes or seconds) – very expensive
- Full Production Scale is running AWS and On Premise



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

All AWS Multi Region



NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

© Stephane Maarek

Disaster Recovery Tips

- Backup
  - EBS Snapshots, RDS automated backups / Snapshots, etc…
  - Regular pushes to S3 / S3 IA / Glacier, Lifecycle Policy, Cross Region Replication
  - From on-premises: Snowball or Storage Gateway
- High Availability
  - Use Route53 to migrate DNS over from Region to Region
  - RDS Multi-AZ, ElastiCache Multi-AZ, EFS, S3
  - Site to Site VPN as a recovery from Direct Connect
- Replication
  - RDS Replication (Cross Region), AWS Aurora + Global Database
  - Database replication from on-premises to RDS
  - Storage Gateway
- Automation
  - CloudFormation / Elastic Beanstalk to re-create a whole new environment
  - Recover / Reboot EC2 instances with CloudWatch if alarms fail
  - AWS Lambda functions for customized automations
- Chaos
  - Netflis has a "simian-army" randomly terminating EC2

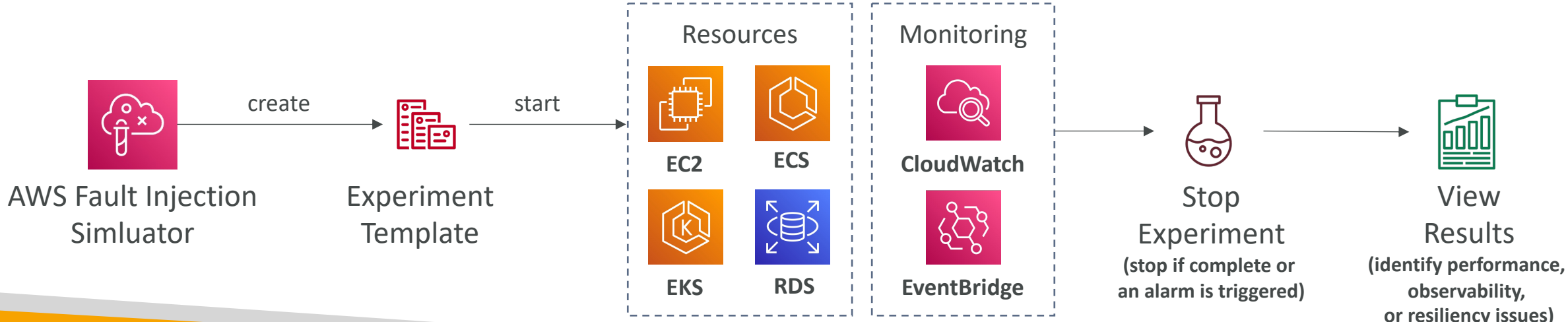
© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS Fault Injection Simulator (FIS)



- A fully managed service for running fault injection experiments on AWS workloads
- Based on **Chaos Engineering** – stressing an application by creating disruptive events (e.g., sudden increase in CPU or memory), observing how the system responds, and implementing improvements
- Helps you uncover hidden bugs and performance bottlenecks
- Supports the following AWS services: EC2, ECS, EKS, RDS…
- Use pre-built templates that generate the desired disruptions



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS Application Discovery Service



- Plan migration projects by gathering information about on-premises data centers
- Server utilization data and dependency mapping are important for migrations
- Agentless Discovery (AWS Agentless Discovery Connector)
  - Open Virtual Appliance (OVA) package that can be deployed to a VMware host
  - VM inventory, configuration, and performance history such as CPU, memory, and disk usage
  - OS agnostic
- Agent-based Discovery (AWS Application Discovery Agent)
  - System configuration, system performance, running processes, and details of the network connections between systems
  - Supports Microsoft Server, Amazon Linux, Ubuntu, RedHat, CentOS, SUSE…
- Resulting data can be exported as CSV or viewed within AWS Migration Hub
- Data can be explorer using pre-defined queries in Amazon Athena

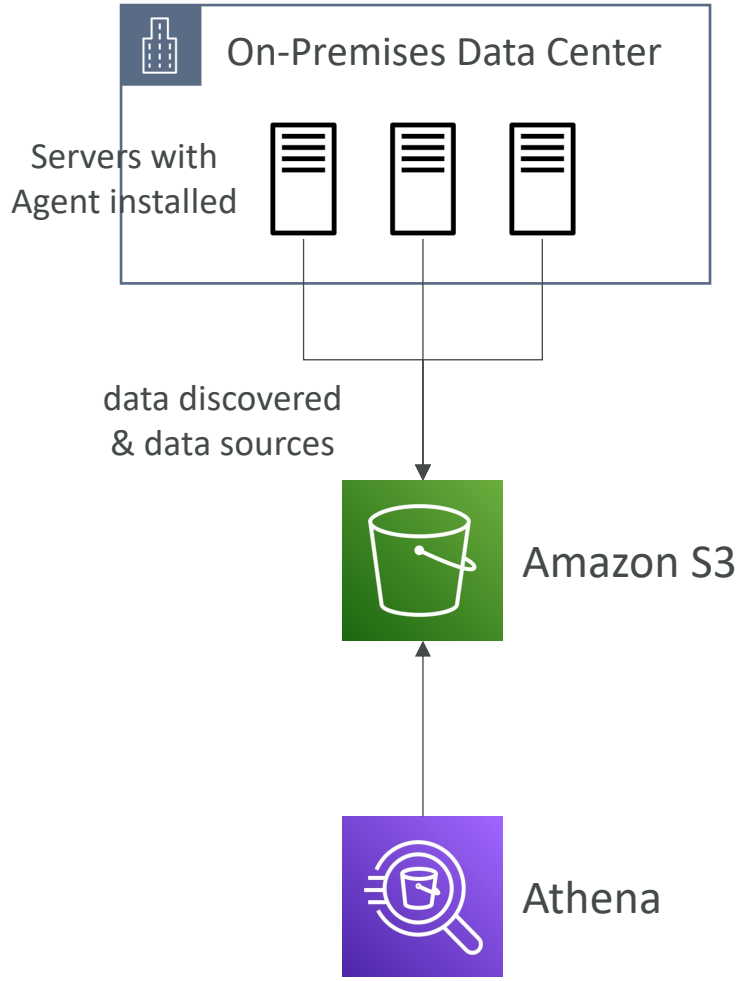
© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS Application Discovery Service – Migration Hub Data Exploration



- Allows you to use Amazon Athena to analyze data collected from on-premises servers during discovery
- Data is automatically stored in S3 bucket at regular intervals
- Use Pre-defined or custom queries in Amazon Athena to analyze data
- Example: type of processes running on each server
- Ability to upload additional data sources such as Configuration Management Database (CMDB) exports
- Integrate Athena with QuickSight to visualize data



© Stephane Maarek

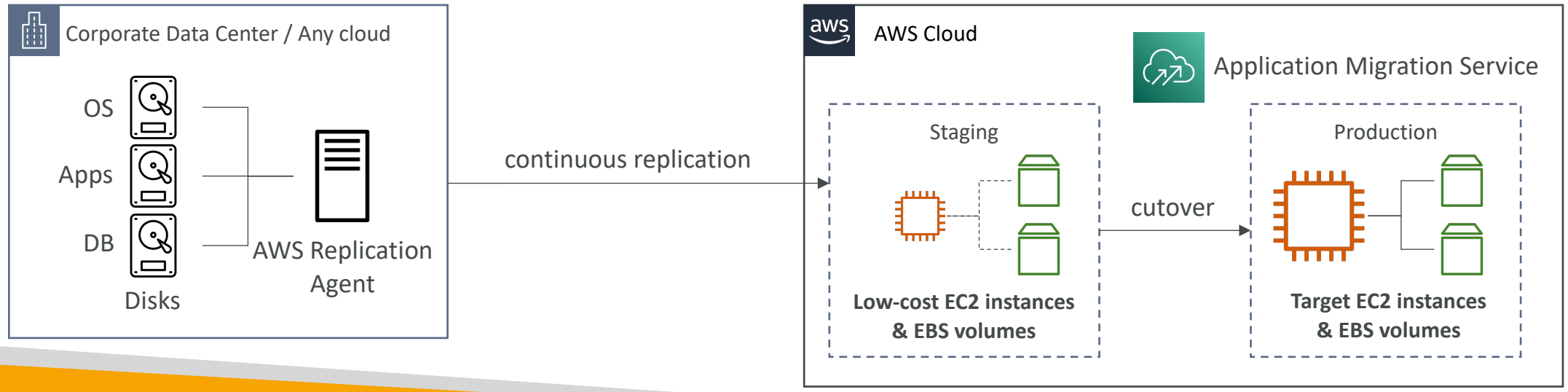
NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS Application Migration Service (MGN)



- *The "AWS evolution" of CloudEndure Migration, replacing AWS Server Migration Service (SMS)*

- Lift-and-shift (rehost) solution which simplify **migrating** applications to AWS
- Converts your physical, virtual, and cloud-based servers to run natively on AWS
- Supports wide range of platforms, Operating Systems, and databases
- Minimal downtime, reduced costs



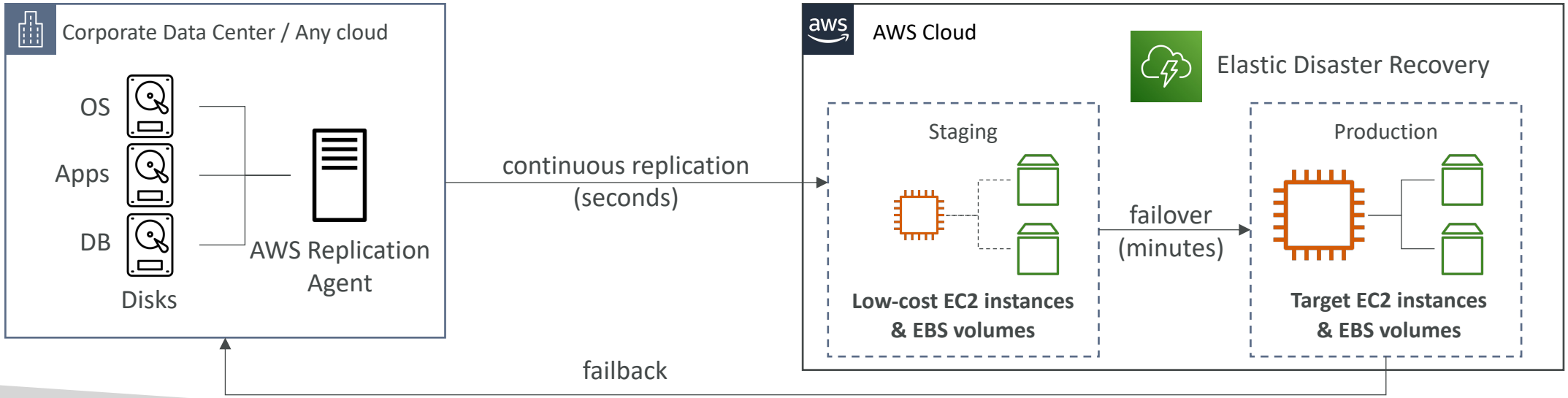
© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS Elastic Disaster Recovery (DRS)

- *Used to be named "CloudEndure Disaster Recovery"*

- Quickly and easily **recover** your physical, virtual, and cloud-based servers into AWS
- Example: protect your most critical databases (including Oracle, MySQL, and SQL Server), enterprise apps (SAP), protect your data from ransomware attacks, …
- Continuous block-level replication for your servers



© Stephane Maarek

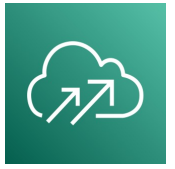
NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

On-premises strategy with AWS

- Ability to download Amazon Linux 2 AMI as a VM (.iso format)
  - VMWare, KVM, VirtualBox (Oracle VM), Microsoft Hyper-V
- AWS Application Discovery Service
  - Gather information about your on-premises servers to plan a migration
  - Server utilization and dependency mappings
  - Track with AWS Migration Hub
- AWS Application Migration Service (MGN)
  - Replacing AWS Server Migration Services & CloudEndure Migration
  - Incremental replication of on-premises live servers to AWS
  - Migrates the entire VM into AWS
- AWS Elastic Disaster Recovery (DRS)
  - Replacing CloudEndure Disaster Recovery
  - Recover on-premises workloads onto AWS
- AWS Database Migration Service (DMS)
  - replicate on-premises => AWS , AWS => AWS, AWS => on-premises
  - Works with various database technologies (Oracle, MySQL, DynamoDB, etc..)



AWS Application Discovery Service



AWS Application Migration Service



AWS Elastic Disaster Recovery



AWS Database Migration Service

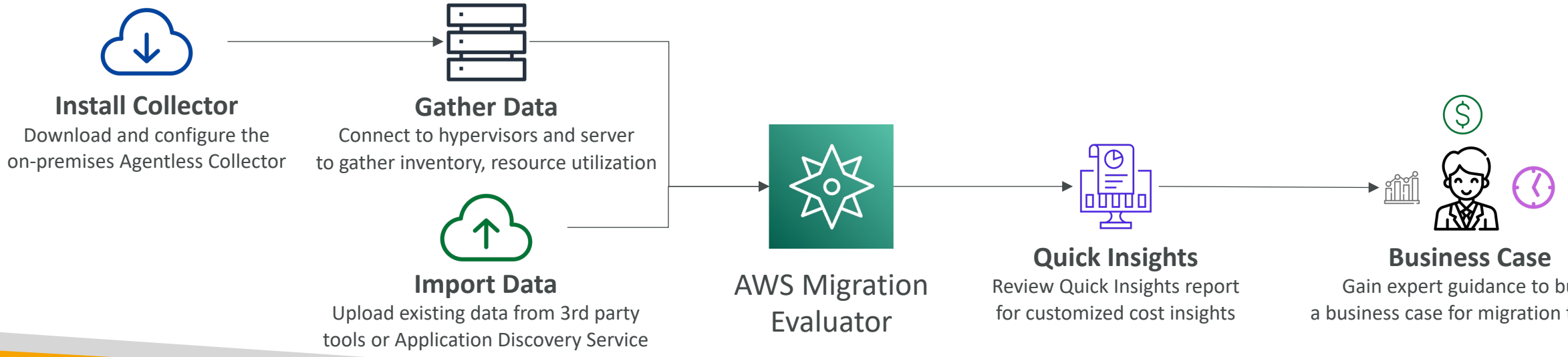
© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS Migration Evaluator



- Helps you build a data-driven business case for migration to AWS
- Provides a clear baseline of what your organization is running today
- Install Agentless Collector to conduct broad-based discovery
- Take a snapshot of on-premises foot-print, server dependepncies, …
- Analyze current state, define target state, then develop migration plan



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS Backup



- Fully managed service
- Centrally manage and automate backups across AWS services
- No need to create custom scripts and manual processes
- Supported services:
  - Amazon EC2 / Amazon EBS
  - Amazon S3
  - Amazon RDS (all DBs engines) / Amazon Aurora / Amazon DynamoDB
  - Amazon DocumentDB / Amazon Neptune
  - Amazon EFS / Amazon FSx (Lustre & Windows File Server)
  - AWS Storage Gateway (Volume Gateway)
- Supports cross-region backups
- Supports cross-account backups

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS Backup

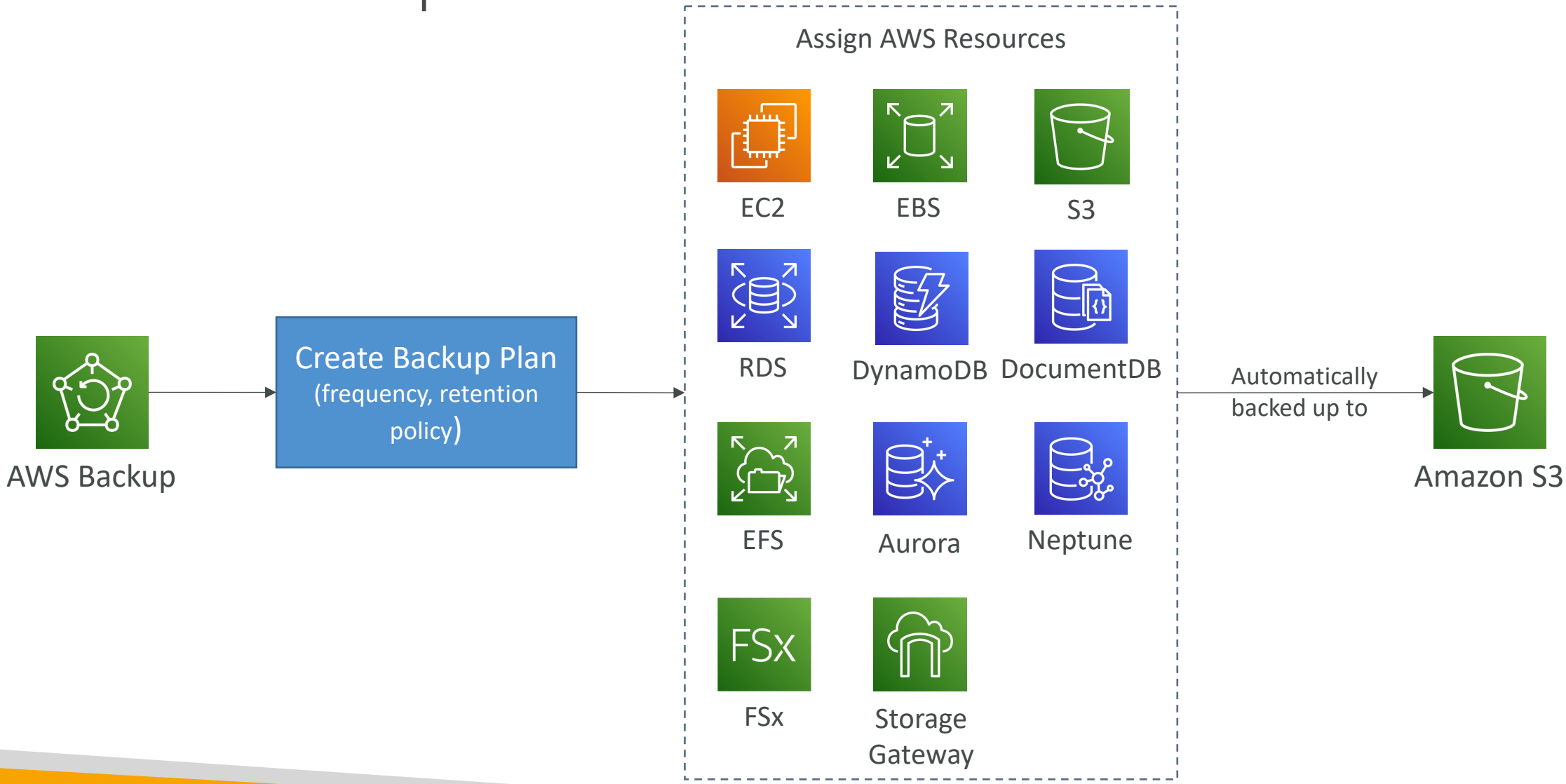


- Supports PITR for supported services
- On-Demand and Scheduled backups
- Tag-based backup policies
- You create backup policies known as **Backup Plans**
  - Backup frequency (every 12 hours, daily, weekly, monthly, cron expression)
  - Backup window
  - Transition to Cold Storage (Never, Days, Weeks, Months, Years)
  - Retention Period (Always, Days, Weeks, Months, Years)

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS Backup

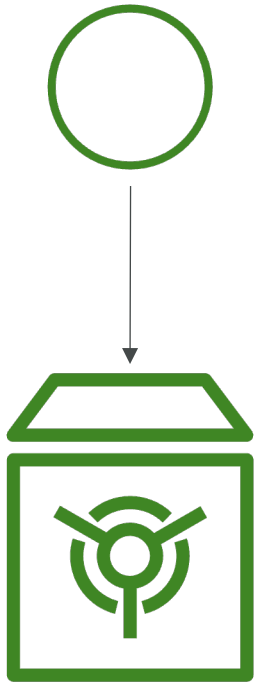


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS Backup Vault Lock

- Enforce a WORM (Write Once Read Many) state for all the backups that you store in your AWS Backup Vault
- Additional layer of defense to protect your backups against:
  - Inadvertent or malicious delete operations
  - Updates that shorten or alter retention periods
- Even the root user cannot delete backups when enabled



backup

Backup Vault Lock Policy
Backups can't be deleted

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com